

Escaping graphical causal priors: Multi-generator cooperation for rationalization

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ABSTRACT

Rationalization is a framework that aims to build self-explanatory NLP models by extracting a subset of human-intelligible pieces of their input texts. It involves a cooperative game where a generator (i.e., selector) selects the most human-intelligible parts of the input as the rationale, followed by a predictor that makes predictions based on these selected rationales. In this study, we aim to equip the rationalization framework with the potential to mine causal rationales from a purely probabilistic perspective, without involving strong assumptions implicit in graphical causal models (e.g., conditional independence and the ignorability of unobservable confounders). The two obstacles that prevent the conventional rationalization framework from selecting causal rationales are spurious correlation and degeneration, where the predictor overfits the spurious or meaningless rationale candidates solely selected by the not-yet well-trained generator and in turn deteriorates the generator. In this paper, we propose a simple yet effective method named MGR (multi-generator based rationalization) to simultaneously solve the two problems, and thus extract the causal rationales. The key idea of MGR is to employ multiple generators such that the occurrence stability of causal parts is improved and more meaningful pieces are delivered to the predictor. Through experiments on six text classification datasets and one graph classification dataset using three network architectures (GRUs, BERT, and GIN), we show that our method significantly outperforms existing rationalization methods. We also compare our method with a representative LLM (LLaMA-3.1-8B-Instruct) and find that our simple method achieves comparable results to those of the LLM.

1. Introduction

With the success of deep learning, there are growing concerns about interpretability [1]. Ideally, the explanation should be both faithful (reflecting the model's actual behavior) and plausible (aligning with human understanding) [2,3]. Post-hoc explanations, which are trained separately from the prediction process, may not faithfully represent the model's decision, despite appearing plausible [1]. In contrast to post-hoc methods, ante-hoc (or self-explaining) techniques typically offer increased transparency [1] and

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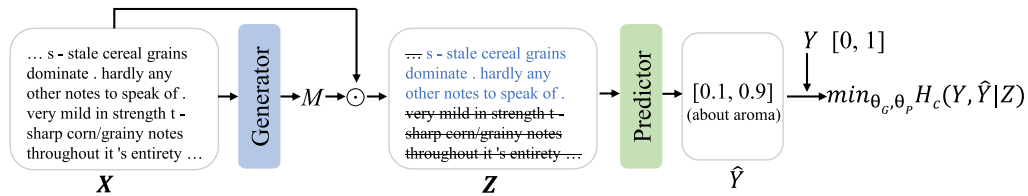


Fig. 1. The standard rationalization framework RNP. The task in this example is binary sentiment classification about beer's aroma. X , Z , $\hat{Y}$, Y represent the input, the selected rationale candidate, the prediction, and the ground-truth label, respectively. M is a sequence of binary masks. θ_G, θ_P represent the parameters of the generator and the predictor, respectively. H_c denotes the cross-entropy.

faithfulness [4], as the prediction is made based on the explanation itself. There is a stream of research that has exposed the unreliability of post-hoc explanations and called for self-explanatory methods [5–8]¹.

In this study, our primary focus is on investigating a general model-agnostic self-explaining framework called Rationalizing Neural Predictions (RNP, also known as rationalization) [9]. Along with its variants, RNP has become one of the mainstream methods for facilitating the interpretability of NLP models [4,10–13], and also holds the potential to be applied to image classification [14] and graph neural networks [15]. RNP utilizes a cooperative game involving a generator and a predictor. This game is designed with a focus on “data-centric” (i.e., it explains the connection between a text and the model-agnostic task label, rather than explaining the output of a specific model) feature importance. The generator first identifies the most informative part of the input text, termed the rationale. Subsequently, the rationale is transmitted to the predictor to make predictions, as illustrated in Fig. 1. The generator and predictor are trained cooperatively to maximize prediction accuracy. Apart from its use for interpretability, some recent studies find that rationalization can also serve as a method for data cleaning. The extracted (Z, Y) pairs can act as a new dataset, and when trained with such a cleaned dataset, a predictor may be more robust [16] and generalizable [17,18], thanks to the removal of task-irrelevant, harmful information. Moreover, these more concise, shorter texts can help reduce computational costs when used with large models [19,20].

Our goal in this study is to enhance this framework to equip it with the ability to discover causal rationales, as causation is more accurate in aiding decision-making than association. Since there are no human-annotated rationales in the training data for supervision, previous methods usually measure the informativeness of the selected rationale through the predictor's prediction accuracy. If the predictor fits spurious features or trivial patterns rather than the causal features, the generator may get incorrect feedback about its selection.

In rationalization, there are two stages where correlations may arise. The first type of correlation originates from the process of data generation, and we refer to it as feature correlation. A typical example of feature correlation, as pointed out in LIME [21], is that wolves often appear together with snow. Consequently, whether the background features contain snow or not can serve as a strong indicator for classifying an image as depicting a wolf. Another instance of feature correlation is demonstrated in Fig. 2(a). Within a beer review, a favorable appearance often correlates with an appealing taste. Comments regarding the taste can serve as a strong indicator for the appearance label. The predictor might inadvertently overfit to such correlations, leading to local optima. Consequently, the suboptimal predictor could mislead the generator to select these spurious correlations. Another type of correlation stems from the rationale (mask) selection stage, and we call it mask correlation. An example of mask correlation is shown in Fig. 2(b). Consider a situation where the explainer has implicitly learned the category of X and selects a “–” for all negative inputs while excluding it from all positive inputs. In this case, the predictor only needs to determine whether the input rationale includes a “–” or not. This phenomenon has not been analyzed from the perspective of spurious correlations in previous literature but has instead been referred to as **degeneration** [22]. However, although it differs from typical spurious correlations (as this correlation does not exist in the original dataset but is introduced by the model), we still classify it as a spurious correlation for the sake of unified analysis and handling.

Some methods have been developed to address either feature correlation or degeneration separately. INVRAT [23] attempts to tackle feature correlation using invariant risk minimization (IRM) [24]. The main idea is to emphasize spurious (non-causal) variations by splitting the dataset into distinct environments. However, IRM-based methods have several limitations. For instance, they require strong prior knowledge about the relationships between non-causal and causal features (e.g., the extra labels of non-causal features) in order to divide the dataset [25]. Moreover, IRM-based methods are limited to addressing only a finite set of predetermined non-causal features, neglecting the potential existence of numerous unknown non-causal features. In fact, a study [25] in the field of IRM has theoretically demonstrated that it is nearly impossible to partition a dataset into different environments to eliminate all non-causal features using IRM. Other challenges, such as the tendency to overfit, difficulty in applying to larger models [26,27], and the marginal shift risk of the input [28], have also been identified in the realm of IRM. Inter_RAT [29] attempts to eliminate feature correlation through *backdoor adjustment*, intervening directly with the confounders. However, it is extremely hard to measure the confounders, since they are usually not observable in the dataset. As for degeneration, although not explicitly associated with spurious correlation until this study, some efforts have been made to fix the problem. The common idea is to introduce regularizers that have access to the full text to regularize the original generator [22,30] or the predictor [4,31]. Regularized by these auxiliary modules, the predictor

¹ Code: <https://github.com/jugechengzi/Rationalization-MGR-VRM>

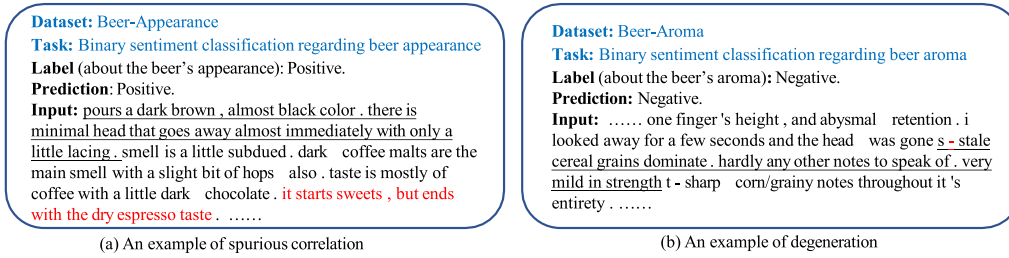


Fig. 2. The underlined text indicates the human-annotated gold-standard rationale, while the red text denotes the rationale selected by RNP. (a) An example of feature correlation (i.e., a typical spurious correlation). This example is drawn from the *Beer-Appearance* dataset, where the task is the binary classification of beer appearance. (b) An example of mask correlation (i.e., degeneration). This example is drawn from the *Beer-Aroma* dataset, where the task is the binary classification of beer aroma. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

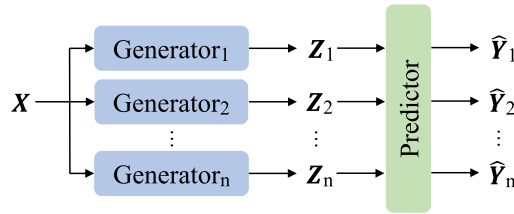


Fig. 3. The architecture of MGR. X is the input full text. Z_i is the selected rationale candidate and $\hat{Y}_i$ is the predictor's output corresponding to Z_i .

can somewhat disregard the mask correlation. However, these methods share a common shortcoming in that the strengths of the regularization terms are difficult to control [13].

In addition to the above-mentioned specific shortcomings of each type of previous methods, they share a common limitation: they are usually designed separately and do not take both problems into account. Moreover, since these methods often introduce complex regularizers (which may lead to conflicts between different approaches), combining two approaches to address the aforementioned problems is also not easy. In this paper, we seek to simultaneously solve the two problems of feature correlation and mask correlation. Specifically, we identify that both problems arise from the fact that the predictor has access only to the limited view of pieces provided by the single generator, and thus may learn incorrectly when this generator selects spurious or meaningless rationales. Besides, previous studies find that the initialization of the model has a significant impact on the training performance, which implicitly indicates that the rationalization model is hard to train once a single generator is not well initialized [4,32].

Considering these limitations of the rationalization with a single generator, as shown in Fig. 3, we design a novel architecture with one predictor but multiple generators. These generators are initialized with different parameters. In this way, the view of the predictor is not limited to a single generator and it can have access to more meaningful rationales. In turn, such a good predictor is then able to give more proper feedback on the informativeness of the selected rationale candidates. We theoretically show that the occurrence stability of real rationales increases such that the predictor has lower risks at overfitting spurious correlations, and that the diversity of rationales is improved such that the predictor can hardly deviate toward a specific meaningless rationale. Experiments conducted on several standard rationalization benchmarks show that MGR (multi-generator based rationalization) achieves significant improvements over several state-of-the-art methods in terms of rationale quality. Our contributions can be summarized as follows:

- Our primary contribution is to provide insights into causal features in the data from a purely probabilistic perspective, without involving graphical causal models. In comparison with methods with graphical causal models, we do not require as many untestable assumptions, such as establishing a structured causal model assuming conditional independence between the variables, assuming the visibility of all confounders for a property of ignorability, or assuming access to interventional data. Note that although it is common to make these assumptions, they are strong assumptions that impose high requirements on the datasets [33,34].
- We propose a simple but effective method, namely MGR, to simultaneously solve the spurious correlations and degeneration problems in rationalization. The key idea is to facilitate the predictor to obtain a broader view of the rationales through multi-generator cooperation.
- We provide theoretical insights that our multi-generator cooperation design can produce real rationales more stably such that the risk of the predictor learning spurious (feature) correlations is reduced. Moreover, we prove that multiple generators can produce more diverse rationales and thus the predictor will not overfit to some specific meaningless rationale, alleviating degeneration (mask correlation).
- Experiments on various rationalization benchmarks show that MGR achieves an improvement of up to 11.9% compared with state-of-the-art rationalization methods in terms of F1 score (we use the overlap between the model-selected and human-annotated rationales to measure the rationale causality).

We note that a shorter conference version of this paper appeared in ACL 2023 [12]. Two limitations remain in the previous conference version: how to reduce model complexity and how to prevent different generators from collapsing. Simultaneously addressing these two issues is challenging. We further extend this method with the idea of vicinal risk minimization [35]. This extended version can prevent different generators from collapsing onto the selection of the same rationale candidates, thus minimizing the risk of overfitting in the predictor. Based on this, we can share the encoder across different generators, reducing the consumption of computational resources. The additional contributions can be summarized as faster and cheaper computation, better performance, and broader applications.

2. Related work

2.1. Self-explaining rationalization

The base cooperative framework of rationalization named RNP [9] is flexible and offers a unique advantage, i.e., certification of exclusion, which means any unselected input is guaranteed to have no contribution to the prediction [4]. However, such a method is challenging to train. To tackle this challenge, many methods have been proposed to improve RNP from various perspectives.

Rationale sampling. Many works focus on refining the sampling process of the rationales. Bao et al. [36] uses Gumbel-Softmax to do the reparameterization of binarized selection. Bastings et al. [37] replaces the Bernoulli sampling distribution with rectified Kumaraswamy distributions. Jain et al. [32] disconnects the training regimes of the generator and predictor via a saliency threshold. Paranjape et al. [38] imposes a discrete bottleneck objective to balance the task performance and the rationale length. Hase et al. [39] explores better metrics for the explanations. Rajagopal et al. [40] uses phrase-based concepts to build a self-explaining model. Other methods, such as employing meta-learning techniques [41], and cooperatively training models using standard continuous and discrete optimization schemes [42], have also been explored. These methods are orthogonal to our method.

Spurious correlations. One important obstacle to rationalization is spurious features in datasets, as these features also have high correlations with the task label and can compete with the causal features for extraction opportunities under the “maximizing prediction accuracy” objective. To tackle this challenge, the typical methods mainly adopt causal inference to penalize the spurious correlations in the original data. For instance, INVRAT [23] attempts to tackle feature correlation using invariant risk minimization (IRM) [24]. The main idea is to penalize spurious (non-causal) variations by splitting the dataset into distinct environments. However, IRM-based methods have several limitations. For instance, they require strong prior knowledge about the relationships between non-causal and causal features (e.g., auxiliary labels for non-causal features) in order to divide the dataset [25]. Moreover, IRM-based methods are limited to addressing only a finite set of predetermined non-causal features, neglecting the potential existence of numerous unknown non-causal features. In fact, a recent study [25] in the field of IRM has theoretically demonstrated that it is nearly impossible to partition a dataset into different environments to eliminate all non-causal features using IRM. Other challenges, such as the tendency to overfit, difficulty in scaling to larger models [26,27], and the marginal shift risk of the input [28], have also been identified within the realm of IRM. Inter_RAT [29] attempts to eliminate feature correlation through *backdoor adjustment*, intervening directly on the confounders. However, it is extremely hard to measure the confounders since they are usually not observable in the dataset. CR [43] calculates the sufficiency and necessity of each token separately, which leads to a high computational complexity, making it feasible only for very short texts. AGR [44] uses reinforcement learning to refine the learning bias, which adds strong assumptions about the unobservable environments ².

Degeneration. Degeneration is one of the major problems that make rationalization frameworks hard to train. To solve this problem, many efforts seek to regularize the predictor using supplementary modules which have access to the information from the full text [4,22,30] such that the generator and the predictor will not collude to produce uninformative rationales. 3PLAYER [22] takes the unselected text Z^c into consideration by inputting it to a supplementary predictor. DMR [30] tries to align the distributions of rationales with the full input text in both the output space and feature space. A2R [4] endows the predictor with full-text information by introducing a soft rationale. GRAT [45] introduces a full-text predictor to match the distributions of the rationale candidates and the full text. However, some recent studies find that additional regularizers can make it challenging to maintain a balance between different modules [4,31]. On the one hand, it makes finding suitable hyperparameters challenging, and on the other hand, these methods may not scale well when combined with other approaches. Yue et al. [46] propose a disentanglement-augmented framework that encourages the model to separate rationale-related explanatory factors from other task-irrelevant or confounding information. Different from them, Noise-RAT [13] tries to avoid overfitting to trivial patterns by injecting noise into the selected rationales.

2.2. Model ensemble methods

Ensemble methods that combine the outputs of several models to improve prediction performance and robustness have been studied for a long time [47–49]. Ensemble methods train N models on different datasets independently and fuse the outputs of different models during inference. This requires maintaining N models and running each of them at test time, which increases the computational costs and brings obstacles to applications [50]. Although our method is similar to ensemble methods to some extent, it has essential differences from ensemble methods. In our method, different generators are not trained entirely independently. In fact, they all play a cooperative game with the same predictor on the same training dataset. With different initializations, different

² We were unable to obtain its code, and therefore did not include it in our experimental comparisons.

generators can provide diverse rationales to train a robust and stable predictor at an early training stage. But with the same training target and dataset, different generators can converge to get the same output [51], thus we only need to keep one generator during inference, which is also empirically supported by the experimental results in Fig. 3(b) of [12].

2.3. The potential impact of rationalization in the era of LLMs

Compared to traditional “model-centric” XAI methods which solely focus on model-learned information, “data-centric” approaches primarily aim to extract model-agnostic patterns inherent to the data. So, apart from improving interpretability, rationalization can serve as a method of data cleaning [52]. Domain-specific large models often require supervised fine-tuning using domain-specific data. Uncleaned data may contain harmful information such as biases and stereotypes [53]. Recent research suggests that training predictors with extracted rationales can remove irrelevant harmful information, enhancing robustness [54] and generalization [17, 18]. Considering that small models are sufficient for simple supervised tasks and are more flexible and cost-effective for training on single datasets (e.g., searching for hyperparameters and adding auxiliary regularizers), using small models for rationalization on a single dataset and then applying the extracted rationales for supervised fine-tuning might prevent large models from learning harmful information from new data. Additionally, shortening input texts can also reduce the memory required for fine-tuning [55]. A recent study also finds that training a small model for data selection and producing a small subset of data is useful for fine-tuning LLMs [56].

3. Task definition

We consider the text classification task, where the input is a text sequence $X = [x_1, x_2, \dots, x_l]$ with x_i being the i -th token and l being the number of tokens. Y represents the class labels in the dataset $\mathcal{D}$. The standard rationalization framework RNP [9] consists of a generator $f_G(\cdot)$ and a predictor $f_P(\cdot)$, with θ_g and θ_p representing the parameters of the generator and predictor. For $(X, Y) \sim \mathcal{D}$, the generator first outputs a sequence of binary masks $M = f_G(X) = [m_1, \dots, m_l] \in \{0, 1\}^l$ (in practice, the generator first outputs a Bernoulli distribution for each token and a mask for each token is independently sampled using Gumbel-Softmax). Then, it forms the rationale candidate Z by the element-wise product of X and M :

$$Z = M \odot X = [m_1 x_1, \dots, m_l x_l]. \quad (1)$$

To simplify the notation, we denote $f_G(X)$ as Z in the following sections, i.e., $f_G(X) = Z$. With the generator’s selection process, we obtain a set of (Z, Y) samples, which are generally considered to follow the distribution $P(Y|Z)$. The rationale Z is searched by maximizing the mutual information $I(Y; Z)$:

$$\begin{aligned} Z^* &= \arg \max_Z I(Y; Z) = \arg \max_Z (H(Y) - H(Y|Z)) = \arg \min_Z H(Y|Z), \\ \text{s.t., } Z &= f_G(X). \end{aligned} \quad (2)$$

In practice, the entropy $H(Y|Z)$ is commonly approximated with the minimum cross-entropy $\min_{\theta_p} H_c(Y, \hat{Y}|Z)$, with $\hat{Y} = f_P(Z)$ representing the output of the predictor. It can be shown that the minimum cross-entropy equals the entropy (see Appendix B.1). Replacing Z with $f_G(X)$, the generator and the predictor are trained cooperatively:

$$\min_{\theta_g, \theta_p} H_c(Y, f_P(f_G(X)) | f_G(X)), \text{ s.t., } (X, Y) \sim \mathcal{D}. \quad (3)$$

To make the selected rationales human-intelligible, rationalization methods usually constrain the rationales with compact and coherent regularization terms. In this paper, we use the most widely used constraints proposed by [23]:

$$\Omega(M) = \lambda_1 \left| \frac{\|M\|_1}{l} - s \right| + \lambda_2 \sum_{i=2}^l |m_i - m_{i-1}|. \quad (4)$$

The first term encourages that the percentage of the tokens being selected as rationales is close to a pre-defined level s . The second term encourages the rationales to be coherent.

4. Multi-generator based rationalization

For the sake of clarity, we first present our approach and then expound on the principles underlying it in Section 5. Based on the framework of RNP, our MGR uses multiple generators with different initialized parameters to help the predictor learn from diverse rationale candidates, as shown in Fig. 3.

Training of MGR. MGR leverages each generator $f_{G_i}(\cdot)$ to process the input text X independently, and then send the obtained rationales Z_i to the predictor $f_P(\cdot)$ to obtain the prediction $\hat{Y}_i$. Based on Eq. (5), MGR computes its loss by calculating the sum of the cross-entropy between Y and $\hat{Y}_i$ in each generator:

$$\min_{\theta_{G_1}, \dots, \theta_{G_n}, \theta_P} \sum_{(X, Y) \sim \mathcal{D}} H(Y, \hat{Y}) = \sum_{(X, Y) \sim \mathcal{D}} \sum_{i=1}^n H(Y, f_P(f_{G_i}(X))). \quad (5)$$

Inference of MGR. During inference, MGR only uses one generator, e.g., the first generator producing Z_1 and $\hat{Y}_1$ in Fig. 3, to provide rationales. It is worth noting that our idea is similar to but not the same as ensemble models. We keep a set of *generators*

Table 1

The payoff (negative loss) table of the cooperative game between the generator and the predictor. Here $a > b$, which indicates that if $f_P(\cdot)$ fits what $f_{G_i}(\cdot)$ selects, both $f_P(\cdot)$ and $f_{G_i}(\cdot)$ get a higher payoff.

$f_{G_i}(\cdot) \backslash f_P(\cdot)$	θ_P^1	θ_P^α	θ_P^0
Select X_1	(a, a)	$(\alpha \cdot a + (1 - \alpha) \cdot b, \alpha \cdot a + (1 - \alpha) \cdot b)$	(b, b)
Select X_2	(b, b)	$((1 - \alpha) \cdot a + \alpha \cdot b, (1 - \alpha) \cdot a + \alpha \cdot b)$	(a, a)

during training to help train a good predictor, which in turn encourages the generators to learn meaningful rationales. Nevertheless, we only keep the first *generator* during inference (see experimental results in Fig. 3(b) of [12]), which is efficient in terms of both time and resource consumption.

5. Theoretical analysis and extended method

As specified in Section 3, the informativeness of the rationale Z during training is measured by the negative cross-entropy through the predictor: $-H_c(Y, \hat{Y}_z|Z)$ (the subscript c denotes cross-entropy, distinguishing it from general entropy). This objective comprises two parts:

$$H_c(Y, \hat{Y}_z|Z) = H(Y|Z) + D_{KL}(P(Y|Z)||P(\hat{Y}_z|Z)), \quad (6)$$

where the entropy $H(Y|Z)$ depends on the dataset itself and the KL divergence represents how well the predictor fits the selected rationale candidate Z . The overall goal of this section is to show how to increase the probability that the predictor fits the causal rationales rather than spurious ones.

5.1. Dealing with spurious correlations

In this subsection, we aim to show that the proposed method can solve the problem of spurious correlations from a purely probabilistic perspective, thus involving no assumptions that are needed for a graphical causal model. The overall idea is to show that the predictor will fit the causal features rather than the spurious features (which in turn will guide the generators to select causal features).

Settings. We consider a simplified setting similar to [4] (we leave the discussion about how the results can generalize to more general cases in Appendix B.8): the input text X consists of two subsets of the features X_1 and X_2 , where X_1 is the causal rationale corresponding to the target aspect label, while X_2 is the comments about other aspects that correlate with the target aspect. We denote the parameters of the predictor by θ_P^1 when it fits X_1 , and by θ_P^0 when it fits X_2 . We also assume a payoff table for the cooperative game between one generator and one predictor, as shown in Table 1, where we denote the negative loss as the payoff. The higher payoff indicates that the rationalization achieves better convergence. Since rationalization is a cooperative game, the payoff of the generator is the same as that of the predictor. Here, a represents the negative loss when the predictor fits the generator's selection and b is the negative loss when the predictor fits the unselected features. Since the loss is usually smaller when $f_P(\cdot)$ fits what $f_{G_i}(\cdot)$ selects, we consider $a > b$. We denote that there are K generators that select the causal rationale X_1 and the other $n - K$ generators select X_2 . Without loss of generality, we consider that the predictor is randomly initialized as an interpolation between θ_P^0, θ_P^1 with θ_P^α , where α is a random variable in $[0, 1]$. Similar to A2R³ [4], we consider α to be the probability (or degree) that the predictor tends to fit X_1 , which means that under θ_P^α , the payoff for selecting X_1 is $\alpha \cdot a + (1 - \alpha) \cdot b$, and $(1 - \alpha) \cdot a + \alpha \cdot b$ for selecting X_2 . If $\alpha = 1$, the predictor always fits X_1 , and if $\alpha = 0$, the predictor always fits X_2 . This corresponds to the third column in Table 1.

In a scenario where K generators select X_1 and the remaining $n - K$ generators select X_2 , with the predictor being θ_P^α , the expected payoff for a predictor is:

$$R_P(\alpha) = K \cdot \alpha \cdot a + K(1 - \alpha) \cdot b + (n - K)(1 - \alpha)a + (n - K) \cdot \alpha \cdot b. \quad (7)$$

We start from the first step of the optimization process: initialization. With the generators' random initialization, each generator follows a Bernoulli distribution $B(1, P_c)$ to select X_1 (or $B(1, 1 - P_c)$ for selecting X_2) (subscript c denotes causality), where P_c depends on the dataset. We assume that $P_c > 0.5$ because the causality appears more often than the spurious correlation when the target aspect's label appears (Please refer to Appendix B.7 and Table B.12 for more detailed discussion). So, K follows a binomial distribution $B(n, P_c)$.

Theoretical analysis.

Lemma 1. For a variable $K \sim B(n, p)$, we have that $E(K) = np$ and $\text{Var}(K) = np(1 - p)$. With the Central Limit Theorem, we have that

$$\frac{K - np}{\sqrt{np(1 - p)}} \xrightarrow{d.f.} X \sim N(0, 1), \text{ s.t. } n \rightarrow \infty. \quad (8)$$

³ Our α is similar to π in [4]. In [4], π is the degree that the generator tends to select X_1 .

Note that [Lemma 1](#) is equal to

$$K \sim N(np, np(1-p)), \text{ s.t. } n \rightarrow \infty. \quad (9)$$

Lemma 2. For a variable $X \sim N(\mu, \sigma^2)$, and an any given constant $t > 0$, we have

$$\Pr(|X - \mu| > t) \leq 2\exp(-\frac{t^2}{2\sigma^2}). \quad (10)$$

The proof is provided in [Appendix B.2](#).

Combining [Lemmas 1](#) and [2](#), we obtain the following property:

Lemma 3. For a variable $K \sim B(n, p)$, if $p > .5$, we have

$$\lim_{n \rightarrow \infty} \Pr(K < n - K) = 0. \quad (11)$$

The proof is provided in [Appendix B.3](#).

Remark 1. This implies that if $P_c > 0.5$ and given a sufficient number of generators, the number of generators that select X_1 will be more than those of selecting X_2 .

Lemma 4. If the number of generators that select X_1 is more than that of those that select X_2 (i.e., $K > n - K$), the predictor will be optimized to increase α to get a higher payoff (i.e., $\frac{\partial R_P(\alpha)}{\partial \alpha} > 0$).

The proof is given in [Appendix B.4](#).

Remark 2. [Lemma 4](#) indicates that if $K > n - K$, by increasing $R_P(\alpha)$ (for lower loss), the cooperative game will guide the predictor to move towards θ_p (by increasing α) and fit the true causal rationale X_1 . In turn, such a refined predictor will guide the generator to select the desired causal rationales, leading to a further increase in K for the next optimization step.

We denote the probability that the predictor overfits the spurious correlation as $\Pr(spu)_n$ (subscript n denotes the number of generators). According to [Lemma 4](#), we have

$$\Pr(spu)_n = \Pr(\frac{\partial R_P(\alpha)}{\partial \alpha} < 0) = \Pr(K < n - K) \quad (12)$$

Combining [Lemmas 4](#) and [3](#), we have the following theorem:

Theorem 1. For any tolerable upper bound on probability P_s of the predictor overfitting to spurious correlations, if $P_c > 0.5$, there exists a constant N such that

$$\forall n > N, \quad \Pr(spu)_n < P_s, \quad (13)$$

$$\forall n > 1, \quad \Pr(spu)_n < \Pr(spu)_{n-1}. \quad (14)$$

The proof is deferred to [Appendix B.5](#).

Remark 3. [Eq. \(13\)](#) indicates that we can reduce the risk of spurious correlations to an arbitrarily low level by increasing the number of generators. [Eq. \(14\)](#) indicates that we can always reduce the risk of spurious correlations by increasing the number of generators. When $n = 1$, MGR becomes the vanilla RNP, and we obtain

$$\Pr(spu)_{RNP} = 1 - P_c. \quad (15)$$

It is obvious that

$$\forall n > 1, \quad \Pr(spu)_{MGR} < \Pr(spu)_{RNP}, \quad (16)$$

which shows that as long as $n \geq 2$, MGR always has lower errors than RNP.

5.2. Dealing with degeneration

As implied in [Section 1](#), degeneration happens in single-generator frameworks because a generator might inadvertently introduce simple trivial patterns to the selected rationale. Compared with genuine semantic features, these trivial patterns, due to their simplicity, might be more easily recognized by the predictor (which is similar to the Simplicity Bias implied in [\[57\]](#)), leading to overfitting on these trivial patterns by the predictor.

Here we will explain why the predictor tends to overfit on these simple patterns. Consider the input X as a set of random variables, or a random vector (not limited to text, text is a specific kind of vector), and Z is a subset of this set. We denote X as $X = \langle Z, X_{-Z} \rangle$, where X_{-Z} is the remaining subset excluding Z . Usually, a predictor f_p tries to fit a distribution $P(Y|A)$ (here A can stand for X or Z) by minimizing the cross-entropy:

$$f_p^* = \arg \min_{f_p} H_c(Y, \hat{Y}|A), \text{ s.t. } \hat{Y} = f_p(A), \quad (17)$$

where the subscript “c” represents cross-entropy, used to differentiate it from joint entropy. We know that $H_c(Y, \hat{Y}|A) = H(Y|A) + D_{KL}(P_{Y|A} || P_{\hat{Y}|A})$, where $H(Y|A)$ is irrelevant to the predictor’s parameters and only depends on the dataset itself. So, the predictor is actually trying to minimize the KL divergence to fit the given distribution.

Then, consider the following lemma:

Lemma 5. Given three random variables $Y, \hat{Y}, A$, where Y and $\hat{Y}$ share the same label set $\mathcal{Y}$, it holds that conditioning on A never leads to a reduction in the divergence:

$$D_{KL}(P_{Y|A}||P_{\hat{Y}|A}) \geq D_{KL}(P_Y||P_{\hat{Y}}). \quad (18)$$

The proof is provided in [Appendix B.6](#).

Remark 4. Here we consider that Z consists of o random variables $Z = \langle Z^1, \dots, Z^o \rangle$, where Z^i is the i -th variable. According to [Lemma 5](#), we have

$$\begin{aligned} D_{KL}(P_{Y|Z}||P_{\hat{Y}|Z}) &= D_{KL}(P_{Y|Z^1, \dots, Z^o}||P_{\hat{Y}|Z^1, \dots, Z^o}) \\ &\geq D_{KL}(P_{Y|Z^i}||P_{\hat{Y}|Z^i}), \\ &\text{s.t. } \forall 1 \leq i \leq o, \end{aligned} \quad (19)$$

which means that simpler patterns are easier to fit for a predictor. If the category information can be trivially reflected by a small subset of Z (i.e., a simple Z^i in (19), e.g., [Fig. 2\(b\)](#)), then it will be easy for the predictor to overfit these trivial patterns rather than fit the true semantics contained in the whole Z , resulting in a local optimum. This problem is somewhat similar to the Simplicity Bias studied in recent paper [\[57\]](#). Notably, we provide a distinct analysis ([Lemma 5](#)) of how this problem arises.

How can multiple generators reduce the risk of overfitting? A well-known method for avoiding overfitting is Vicinal Risk Minimization (VRM) [\[35\]](#). The central concept of VRM is to compel the predictor to perform well in the vicinity of the original data samples, thereby preventing the predictor from relying solely on memorization of a limited set of samples for making predictions. The core challenge for VRM is how to construct vicinal samples. A method that achieves great success in the field of computer vision is mix-up [\[58\]](#), which produces vicinal samples by performing linear interpolation between pairs of positive and negative samples. However, this is not feasible for natural language, as the interpolated natural language lacks meaningful context. However, we find that using multiple generators has the potential to produce meaningful vicinal samples for language (note that the term “samples” here refers to the predictor’s input Z , i.e., the rationale candidates, rather than the full input X). Consider that the full input X consists of the true rationale R and some noise N : $X = \langle R, N \rangle$. For the i -th generator’s output $Z_i = f_{G_i}(X)$, we can denote it as $Z_i = \langle \psi_i(R), \psi_i(N) \rangle$. With multiple different ψ_i , we have two advantages: Firstly, different $\psi_i(R)$ may recover the semantic components of R missed during individual sampling. Secondly, different $\psi_i(N)$ can introduce different noise patterns, effectively building a series of vicinal samples around a single sampling point, reducing the risk of overfitting to a specific noise.

Another common way to avoid overfitting is to improve the diversity of the training data. In the context of rationalization, we can improve the diversity of sampled rationale candidates Z , i.e., $H(Z)$, to avoid predictor overfitting. Next, we will show that MGR with multiple generators improves the diversity of the selected rationale candidates. Specifically, by treating the rationales of different generators as different variables, we can compute the rationale diversity of MGR as

$$H(Z_{MGR}) = H(Z_1, Z_2, \dots, Z_n). \quad (20)$$

Theorem 2. For $\forall i \in [1, n]$, we have

$$H(Z_i) \leq H(Z_1, Z_2, \dots, Z_n) \leq \sum_{k=1}^n H(Z_k), \quad (21)$$

where the equality on the right holds if and only if $\forall i, j, Z_i \perp Z_j$, and the equality on the left holds if and only if $\forall i, j, Z_i = Z_j$.

The proof is given in [Appendix B.9](#). [Theorem 2](#) indicates that the diversity of MGR with multiple generators is equivalent to the case with a single generator when all generators are the same. More specifically, since RNP consists of only one generator, we have

$$H(Z_{RNP}) = H(Z_i) \leq \max_i H(Z_i), \quad (22)$$

where $i \in [1, n]$. We always have $H(Z_{MGR}) \geq H(Z_{RNP})$ regardless of how the generators are coupled, thereby alleviating degeneration.

In addition, [Theorem 2](#) also indicates that the diversity of MGR achieves the maximum when all the generators are completely independent. Accordingly, we seek to decrease the correlation between different generators to make $H(Z_{MGR})$ gets closer to $\sum_{k=1}^n H(Z_k)$ during training, which will be specified in the next section.

Intuitive understanding. Here we provide a more intuitive understanding of the effectiveness of MGR in dealing with degeneration. Degeneration can be understood as collusion between a generator and a predictor: the generator adds some shortcuts to the selected rationale candidates and the predictor learns these shortcuts. We borrow the idea from the invariant risk minimization (IRM) [\[24\]](#) philosophy. In IRM, a predictor that performs well on several datasets collected under different environments is thought to be able to ignore the shortcuts, since the shortcuts vary across different environments. The generators in our MGR play a similar role as the environments in IRM: The predictor’s training data (i.e., the rationale candidates) are collected by different generators. If the generators are diverse, the shortcuts added by different generators may also vary, similar to those in different environments.

5.3. Extending MGR with vicinal risk minimization

The above analysis has shown that VRM may be helpful in alleviating degeneration. In this section, we provide a practical solution by applying VRM to MGR.

Combining (4) and (5), the overall objective of MGR is

$$\min_{\theta_{G_1}, \dots, \theta_{G_n}, \theta_P} \sum_{(X,Y) \sim \mathcal{D}} \sum_{i=1}^n (H_c(Y, f_P(M_i \odot X)) + \Omega(M_i)), \quad s.t. \quad M_i = f_{G_i}(X). \quad (23)$$

We denote $M_i = [m_1^i, \dots, m_l^i]$, where l denotes the sequence length. $\Omega(M_i)$ is the shortness and coherence regularizer for the rationale candidates selected by the i -th generator and it depends only on θ_{G_i} , and is irrelevant to the predictor. Instead of directly forcing the generators' parameters to diverge from each other (which may influence the convergence point of the cooperative games between the predictor and the generators, i.e., $H_c(Y, f_P(M_i \odot X))$), we make a slight modification to $\Omega(M_i)$ to keep different generators' selecting diverse rationale candidates.

$$\Omega(M_i) = \lambda_1 \left| \frac{\|M_i\|_1}{l} - s_i \right| + \lambda_2 \sum_{t=2}^l |m_t^i - m_{t-1}^i|. \quad (24)$$

We make a very simple modification to the original MGR: we set different s_i for different generators. That is to say, we force different generators to select rationales with different sparsity. This extended version is called MGR-VRM.

The advantages of this trick can be understood from two perspectives. First, its effectiveness. Second, its flexibility in usage. By adding different sparsity constraints to different generators, the selected rationale candidates will cover a wider vicinity (note that the term "vicinity" here is about the predictor's input space Z , i.e., the rationale candidates, rather than the full input X), thus we can achieve a more comprehensive sampling within the vicinity, thereby better utilizing VRM to alleviate degeneration. Simultaneously, this can further improve $H(Z)$ (as specified in Section 5.2). From a usage perspective, different situations (e.g., users) may require different explanation complexities (i.e., rationale lengths). For instance, laypersons might need simpler explanations for an intuitive understanding of the model's predictions. On the other hand, experts may require more detailed explanations to assess whether the model is functioning properly. This happens commonly in federated learning, where users vary significantly. Our MGR-VRM can provide rationales with different lengths, thus it does not require retraining once the length requirement changes.

In practice, if the encoders in different generators are shared (thus reducing the computational cost), the original MGR may suffer significant performance degradation (see Fig. 4 of [12]), because the original MGR cannot guarantee that the different generators sample diverse rationales. But by assigning different s_i to different generators, the generators are expected to sample rationale candidates from different vicinities. Based on this property, we now can share the encoders across different generators, thereby significantly saving computational resources. The architecture of MGR-VRM is shown in Fig. 4.

6. Experiments

6.1. Experimental setup

To begin with, we note that our primary goal is not to improve out-of-distribution prediction performance. Instead, our main focus is to extract causal rationales to improve human understanding (i.e., interpretability).

Baselines. In practice, we follow the vanilla MGR [12] to set $n = 3$ (the number of generators) for the extended MGR-VRM, as previous experiments on MGR have shown that $n = 3$ is a good performance-resource trade-off. We compare MGR-VRM to the standard RNP [9] and the vanilla MGR [12]. We also include several recently published methods that aim to address either spurious correlations or degeneration. INVRAT [23] and Inter-RAT [29] are designed to address spurious correlations, and Noise-RAT [13] and GRAT [45] are designed to address degeneration. All of the baselines have been specified in Section 2. Aside from these methods, we also compare with two well-known traditional post-hoc methods, LIME [21] and Attention (the version of [9]).

Implementation details. Since some of the previous baselines are highly sensitive to hyperparameters and require meticulous hyperparameter tuning, we adopt the same main settings as prior baselines to reduce the difficulty of hyperparameter search and thus avoid unfair comparisons caused by hyperparameters. However, since different baselines in previous work did not fully adopt a unified configuration, we first follow INVRAT, Inter-RAT, and GRAT to conduct experiments with 100-dimensional GloVe [59] for word embedding and 200-dimensional GRUs [60] for text representation. For the main experiments, we follow INVRAT and Inter-RAT

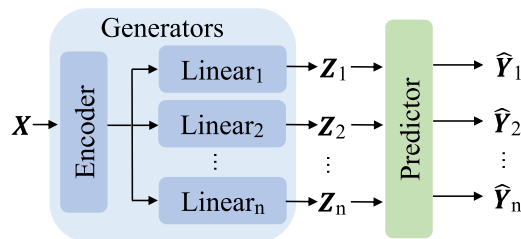


Fig. 4. The architecture of MGR-VRM. Each generator consists of an encoder (e.g., RNN or Transformer) and a linear layer. So does the predictor. The encoders of different generators are shared to save the computational resources.

to take three different rationale sparsity levels: 10%, 20%, 30%. Then, we follow CR to conduct experiments with the pre-trained BERT model (“bert-base-uncased” [61]) as a supplement. Finally, to verify the generalizability of our findings, we extend our method to the graph domain and perform experiments on graph data using GIN. We also compare with a representative LLM, LLaMA3.1-8B-Instruct, with both supervised fine-tuning and few-shot prompting. We use Adam [62] as the optimizer. All of the models are trained on an RTX 3090 GPU. More details are in [Appendix A.1](#).

Datasets. We conduct experiments on six text classification datasets and one graph classification dataset. The first three datasets are collected from the BeerAdvocate benchmark [63]. They are: Beer-Appearance, Beer-Aroma, and Beer-Palate. These datasets contain the binary classification labels for beer appearance, aroma, and palate, respectively. They were originally collected by Lei et al. [9] from the BeerAdvocate platform. The next three datasets are collected from the HotelReviews [64] benchmark. They are Hotel-Location, Hotel-Service, Hotel-Cleanliness. These datasets contain the binary classification labels for hotel location, service, and cleanliness, respectively. They were originally collected by [37] from HotelReviews [64].

In the above datasets, each text contains comments discussing different aspects of beer/hotel. The comments about different aspects are highly correlated but the classification label is only caused by the comment about the target aspect. These properties make the above datasets appropriate for verifying models’ ability for causal discovery. Among them, the three beer-related datasets are most commonly used in the field of rationalization, and we take them as the main datasets. The hotel-related datasets serve as a supplement.

As for the graph dataset, we consider the GOODMotif dataset provided by [65]. This is a synthetically generated graph classification dataset, which contains severe spurious correlations.

All of the above datasets contain ground-truth rationales in their test sets, which are used to objectively compare the performance of different methods. However, in most cases, the training sets do not include manually annotated rationales.

Metrics. Due to the existence of spurious correlations, predictive performance is not a good metric for rationale causality. Since annotators assign the label of the target aspect by observing the causal features, the overlap between the model-selected tokens and human-annotated rationales provides a more direct metric for rationale causality. The terms P , R , and F_1 denote precision, recall, and F_1 score, respectively. These metrics are the most frequently used in rationalization. The term S represents the average sparsity of the selected rationales, that is, the percentage of selected tokens relative to the full text. Following the most relevant baseline Inter-RAT [29], we set the sparsity approximately 10%, 20% and 30%, by adjusting s_i in (24). In short, we use the same metrics as Inter-RAT. Since our MGR-VRM has 3 different generator heads, we set S_i to approximately $i \times 10\%$, such that we can obtain the results of three sparsity levels within a single training run.

6.2. Results

Rationale causality. The results on the beer-related and hotel-related datasets are shown in [Tables 2](#) and [3](#), respectively. The primary metric is the F_1 score. The best results are shown in **bold**. The second-best F_1 scores of previous methods are underlined. We see that, in most cases, vanilla MGR already outperforms other baselines in terms of the F_1 score. The extended version MGR-VRM can even perform better than vanilla MGR. A visualized example is provided in [Appendix A.3](#). Another thing that needs to be noted is that MGR-VRM has much lower computational cost as compared to vanilla MGR (due to the sharing of different generators’ encoders, further illustrated in [Fig. 5](#) later).

Another observation is that, generally, previous methods achieve poor results when the sparsity is low but get comparable results to MGR when the sparsity is high. The possible reason is that, when the sparsity is low ($S \approx 10\%$), non-causal methods may randomly select either the spuriously correlated features (e.g., X_2 described in the beginning of [Section 5.1](#)) or the causal feature (e.g., X_1 described in [Section 5.1](#)); while when the sparsity is high ($S \approx 30\%$), both causal and non-causal methods will select the spurious feature and the causal feature simultaneously, thus achieving similar results. These results further validate the reasonableness of the [Section 5.1](#) settings to some extent.

From a data cleaning perspective, we can observe a general tendency for recall to increase as the sparsity increases. We see that when the sparsity is about 30%, the recall of MGR-VRM is more than 95% on the Beer-Appearance dataset, which means that almost all task-relevant causal rationales have been preserved with only 30% of the original text length. Note that in addition to removing malicious information, a smaller text length can significantly reduce GPU memory usage when it is used to fine-tune large models [19,20].

Results with BERT encoder. Due to the sensitivity of some baselines to hyperparameters and the exacerbation of this sensitivity when using an over-parameterized BERT, we do not use BERT in the main experiments but keep the settings consistent with previous baselines. However, we conduct supplementary experiments on the most widely used Beer-Appearance dataset using BERT as the encoder for both the generator and the predictor. We compare the results with some methods that have been implemented with BERT in the previous work. The results are shown in [Table 4](#). We follow CR [43] to report the results with the sparsity set to approximately 10% (although we still train our MGR-VRM with three sparsity levels in a single run). We still outperform the previous SOTA implemented with BERT by a large margin, showing the generalization of the proposed MGR-VRM.

Saving the computational costs. Aside from the performance, one of the most important advantages of MGR-VRM over the vanilla MGR is the lower computational cost. In the original MGR [12], sharing encoders between different generators leads to decreased performance. However, MGR-VRM achieves comparable performance to MGR, with the shared encoders. In [Fig. 5](#), we show the difference between sharing and not sharing encoders in terms of the training time and GPU memory. MGR-VRM can save a lot of computational resources by sharing the encoders (especially when the batch size is large), while it does not decrease performance. Notably, our MGR-VRM has achieved comparable time efficiency with the vanilla RNP and comparable memory usage with the latest

Table 2

Results on datasets from the BeerAdvocate benchmark (averaged with five random seeds). “*”: results obtained from Inter-RAT [29].

Methods		Beer-Appearance				Beer-Aroma				Beer-Palate			
		S	P	R	F1	S	P	R	F1	S	P	R	F1
$S \approx 10\%$	LIME	10.1	58.5	31.9	41.3	10.1	34.0	22.0	26.7	10.1	5.4	4.2	4.9
	Attention	9.6	65.2	33.9	44.7	9.6	58.9	36.4	45.0	9.6	43.3	33.6	37.9
	RNP*	10.0	32.2	18.6	23.6	10.0	44.8	32.4	37.6	10.0	24.6	23.5	24.0
	INVRAT*	10.0	42.6	31.5	36.2	10.0	41.2	39.1	40.1	10.0	34.9	45.6	39.5
	Inter-RAT*	11.7	66.0	46.5	54.6	11.7	55.4	47.5	51.1	12.6	34.6	48.2	40.2
	Noise-RAT	11.6	72.2	45.3	55.7	10.7	67.6	46.3	55.0	11.0	61.4	54.0	<u>57.5</u>
	GRAT	10.0	88.4	48.0	<u>62.2</u>	10.4	87.0	52.9	<u>65.8</u>	10.4	42.6	33.2	37.3
	MGR	10.9	87.5	51.7	65.0	10.3	78.7	52.2	62.8	10.8	65.6	57.1	61.1
$S \approx 20\%$	MGR-VRM	10.5	91.7	51.8	66.2	9.6	90.5	55.5	68.8	11.9	63.8	60.8	62.2
	LIME	20.1	49.3	53.6	51.3	20.1	31.7	40.9	35.7	20.1	5.3	8.6	6.6
	Attention	19.7	52.6	56.0	54.2	19.7	46.1	58.3	51.5	19.7	34.0	53.8	41.7
	RNP*	20.0	39.4	44.9	42.0	20.0	37.5	51.9	43.5	20.0	21.6	38.9	27.8
	INVRAT*	20.0	58.9	67.2	62.8	20.0	29.3	52.1	37.5	20.0	24.0	55.2	33.5
	Inter-RAT*	21.7	62.0	76.7	68.6	20.4	44.2	65.4	52.8	20.8	26.3	59.1	36.4
	Noise-RAT	20.2	70.1	76.4	73.1	21.1	55.7	75.3	<u>64.0</u>	18.9	41.0	62.5	49.5
	GRAT	20.3	76.4	79.0	<u>77.7</u>	20.3	56.4	68.1	61.7	20.1	43.6	66.4	<u>52.6</u>
$S \approx 30\%$	MGR	20.3	76.3	83.6	79.8	19.7	64.4	81.3	71.9	19.3	47.1	73.1	57.3
	MGR-VRM	20.3	77.3	84.9	80.9	19.2	69.3	85.6	76.6	19.7	47.3	74.8	57.9
	LIME	30.1	42.5	69.2	52.7	30.1	29.8	57.5	39.2	30.1	5.4	13.1	7.7
	Attention	29.6	43.7	69.9	53.8	29.6	37.9	72.0	49.6	29.6	27.8	66.2	39.1
	RNP*	30.0	24.2	41.2	30.5	30.0	27.1	55.7	36.4	30.0	15.4	42.2	22.6
	INVRAT*	30.0	41.5	74.8	53.4	30.0	22.8	65.1	33.8	30.0	20.9	71.6	32.3
	Inter-RAT*	30.5	48.1	82.7	60.8	29.4	37.9	72.0	49.6	30.4	21.8	66.1	32.8
	Noise-RAT	29.6	59.0	94.4	<u>72.6</u>	89.5	44.2	84.5	<u>58.1</u>	28.7	24.8	57.3	34.6
$S \approx 40\%$	GRAT	30.3	59.1	92.5	72.2	30.3	36.7	67.6	47.6	30.3	33.5	77.3	<u>46.8</u>
	MGR	30.4	57.2	93.9	71.1	29.8	45.8	87.4	60.1	30.3	27.3	66.5	38.7
	MGR-VRM	29.3	60.7	96.1	74.4	29.4	48.8	92.3	63.9	29.0	35.5	82.8	49.7

We omit the standard deviation to prevent the table from becoming too wide or wrapping.

Table 3

Results on datasets from the HotelReviews benchmark.

Methods		Hotel-Location				Hotel-Service				Hotel-Cleanliness			
		S	P	R	F1	S	P	R	F1	S	P	R	F1
$S \approx 10\%$	LIME	10.1	13.1	15.7	14.3	10.1	12.3	10.8	11.5	10.1	11.0	12.6	11.7
	Attention	9.7	18.5	21.1	19.7	9.7	15.9	13.5	14.6	9.7	19.5	21.3	20.4
	RNP	9.0	46.3	49.4	47.8	9.0	36.5	28.5	32.0	10.5	26.4	31.4	<u>28.7</u>
	Inter-RAT	11.0	34.7	44.8	39.1	11.9	36.5	38.5	37.5	10.2	22.2	25.5	23.7
	Noise-RAT	8.9	41.0	43.2	42.1	8.7	31.6	24.0	27.3	9.9	26.1	29.5	27.9
	GRAT	9.6	47.1	55.7	<u>51.0</u>	10.2	48.9	36.9	<u>42.1</u>	10.4	18.2	18.7	18.5
	MGR	8.8	54.2	56.1	55.1	10.6	45.6	42.1	43.8	10.5	37.5	44.4	40.6
	MGR-VRM	8.5	56.5	56.3	56.4	8.8	51.9	40.0	45.2	10.4	35.2	41.2	38.0
$S \approx 20\%$	LIME	20.2	11.3	27.0	16.0	20.2	12.2	21.4	15.5	20.1	10.5	23.9	14.6
	Attention	19.7	13.0	30.3	18.2	19.8	13.3	22.9	16.8	19.8	14.9	33.2	20.5
	RNP	20.1	33.2	78.7	<u>46.7</u>	19.5	28.5	49.1	36.1	21.3	22.3	53.6	31.5
	Inter-RAT	21.2	26.2	65.5	37.4	20.2	29.6	32.3	37.8	19.9	20.5	46.1	28.4
	Noise-RAT	19.8	32.8	76.6	46.0	19.8	23.4	40.5	29.7	19.9	22.7	51.2	<u>31.5</u>
	GRAT	20.1	24.1	52.1	33.0	20.2	23.2	38.8	29.0	21.5	16.8	37.0	23.1
	MGR	19.5	34.7	79.7	48.3	20.7	30.2	54.4	38.8	20.6	24.1	56.1	33.7
	MGR-VRM	18.8	35.9	79.7	49.5	19.8	33.8	58.4	42.9	20.2	24.8	56.6	34.5
$S \approx 30\%$	LIME	30.2	10.5	37.2	16.3	30.2	11.9	31.4	17.3	30.2	9.9	33.7	15.3
	Attention	29.7	10.8	37.7	16.7	29.7	13.0	33.7	18.8	29.7	15.2	50.9	23.4
	RNP	29.7	23.3	81.6	36.3	30.3	25.3	66.9	36.7	31.0	17.5	61.3	27.2
	Inter-RAT	29.1	20.7	70.8	32.0	30.3	24.7	65.3	35.9	30.9	14.7	51.4	22.9
	Noise-RAT	29.9	24.1	85.0	<u>37.6</u>	29.1	25.8	65.5	<u>37.0</u>	30.5	18.3	63.3	<u>28.5</u>
	GRAT	30.5	19.0	64.3	29.3	30.1	21.9	56.0	31.5	31.2	14.8	50.0	22.8
	MGR	29.7	23.5	82.3	36.6	30.7	23.9	64.1	34.8	30.9	18.6	64.9	28.9
	MGR-VRM	28.5	26.5	88.8	40.8	29.6	26.8	69.4	38.7	29.8	19.1	64.1	29.4

We omit the standard deviation to prevent the table from becoming too wide or wrapping.

Table 4
Results with BERT encoder on *Beer-Appearance* dataset.
“*”: results obtained from [43].

Methods	Beer-Appearance			
	S	P	R	F1
RNP	10.0 $\pm$ 0.5	47.9 $\pm$ 33.6	20.6 $\pm$ 15.4	28.8 $\pm$ 21.2
RNP*	10.0 $\pm$ 0.0	40.0 $\pm$ 1.4	20.3 $\pm$ 1.9	25.2 $\pm$ 1.7
VIB*	10.0 $\pm$ 0.0	52.6 $\pm$ 2.0	26.0 $\pm$ 2.3	32.9 $\pm$ 2.1
A2R*	10.0 $\pm$ 0.0	55.0 $\pm$ 0.8	25.8 $\pm$ 1.6	34.3 $\pm$ 1.4
CR*	10.0 $\pm$ 0.0	59.7 $\pm$ 1.9	31.6 $\pm$ 1.6	39.0 $\pm$ 1.5
MGR-VRM	11.3 $\pm$ 0.2	82.0$\pm$3.2	38.9$\pm$2.0	52.8$\pm$2.5

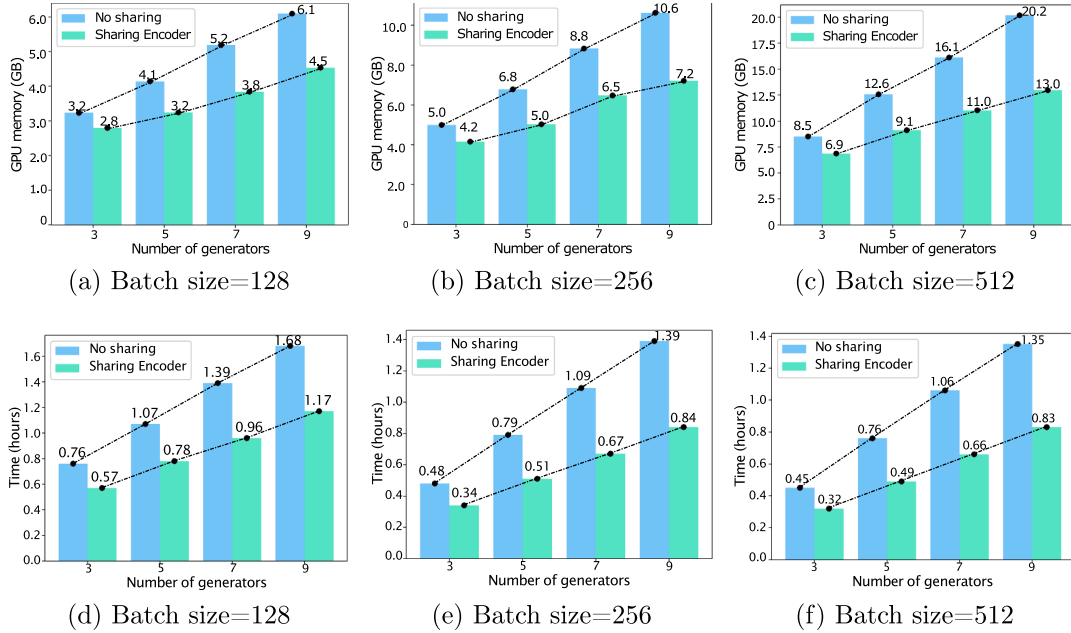


Fig. 5. The comparison of computational cost between sharing and no sharing encoders. (a)(b)(c): the memory usage (MB). (d)(e)(f): the training time (RTX 3090 hours). The time is about training 300 epochs on *Beer-Appearance*.

causal method, Inter-RAT, as shown in Table 5. It is worth noting that our MGR-VRM can achieve three sparsity levels within a single training run, whereas other methods require three training sessions. Therefore, the time efficiency of our MGR-VRM will be higher than that shown in Table 5.

Experiments with graph data. To show the generalizability of our findings to different domains, we further extend MGR to the graph domain. We conduct experiments with the GOODMotif dataset [65] (it is a three-class graph classification dataset with manually injected spurious correlations, and it also has ground-truth rationale labels). For graph data, we use a three-layer GIN as the semantic encoder. Since most rationalization methods are designed specifically for text data, they cannot be applied to the graph domain. The only two approximate baselines are RNP and GRAT [45]. The results are in Table 7. We see that our MGR-VRM still performs well on graph datasets.

Comparison with LLMs. To further show the potential impact of rationalization in the era of LLMs, we present experimental results using a representative LLM, LLaMA-3.1-8B-Instruct. We perform both two-shot prompting and supervised fine-tuning. We do not include GPT-4 in the comparison because GPT-4 requires extensive human alignment, which is very expensive. GPT-4 is overly

Table 5
Time and memory of different methods. The time is measured based on the optimal parameters for each method, and the memory usage is measured based on a uniform batch size of 128.

	RNP	Noise-RAT	Inter-RAT	MGR	MGR-VRM
Time (RTX3090 hours)	0.44	0.50	2.21	0.76	0.57
Memory (MB)	2186	2478	2558	3230	2794

Table 6

The comparison between our MGR-VRM and a representative LLM LLaMA-3.1-8B-Instruct.

(a) Results on datasets from the BeerAdvocate benchmark.												
Methods \ Datasets	Beer-Appearance				Beer-Aroma				Beer-Palate			
	S	P	R	F1	S	P	R	F1	S	P	R	F1
llama-3.1-8b (finetune)	9.9	86.3	46.2	60.2	10.8	73.2	50.6	59.8	8.6	61.7	42.6	50.4
llama-3.1-8b (2 shot)	19.2	15.4	16.0	15.7	21.1	17.9	24.2	20.6	21.2	13.0	22.2	16.4
MGR-VRM	10.5	91.7	51.8	66.2	9.6	90.5	55.5	68.8	11.9	63.8	60.8	62.2
	20.3	77.3	84.9	80.9	19.2	69.3	85.6	76.6	19.7	47.3	74.8	57.9
	29.3	60.7	96.1	74.4	29.4	48.8	92.3	63.9	29.0	35.5	82.8	49.7
(b) Results on datasets from the HotelReview benchmark.												
Methods \ Datasets	Hotel-Location				Hotel-Service				Hotel-Cleanliness			
	S	P	R	F1	S	P	R	F1	S	P	R	F1
llama-3.1-8b (finetune)	5.7	58.6	39.0	46.8	6.0	77.3	40.6	53.3	5.1	54.9	31.3	39.9
llama-3.1-8b (2 shot)	11.0	45.8	59.1	51.6	13.1	45.3	51.7	48.3	9.7	39.3	43.0	41.1
MGR-VRM	6.0	59.2	39.8	47.6	6.1	76.5	41.2	53.5	5.3	55.1	31.9	40.4
	8.5	56.5	56.3	56.4	8.8	51.9	40.0	45.2	10.4	35.2	41.2	38.0
	18.8	35.9	79.7	49.5	19.8	33.8	58.4	42.9	20.2	24.8	56.6	34.5
	28.5	26.5	88.8	4.08	29.6	26.8	69.4	38.7	29.8	19.1	64.1	29.4

Table 7

Results with on the graph domain dataset GOODMotif. The sparsity is set to about 30% (close to the sparsity of ground-truth rationales).

Methods	GOODMotif			
	S	P	R	F1
RNP	31.0 $_{\pm 1.3}$	43.0 $_{\pm 2.9}$	45.3 $_{\pm 2.1}$	44.1 $_{\pm 2.3}$
GRAT	31.5 $_{\pm 2.1}$	43.6 $_{\pm 3.7}$	46.7 $_{\pm 3.9}$	45.1 $_{\pm 3.8}$
MGR	30.4 $_{\pm 0.9}$	45.8 $_{\pm 1.9}$	47.4 $_{\pm 2.3}$	46.6 $_{\pm 2.1}$
MGR-VRM	31.0 $_{\pm 1.1}$	47.4 $_{\pm 1.7}$	50.0 $_{\pm 2.4}$	48.6 $_{\pm 2.1}$

powerful and therefore not representative, as many studies even use GPT-4 as an evaluator to evaluate different methods. Additionally, GPT-4 cannot be easily deployed privately.

For two-shot prompting, we provide the model with one negative text with its corresponding rationale, and one positive text with its corresponding rationale. For supervised fine-tuning, the supervision signal is the classification label, as we perform unsupervised rationale extraction. More implementation details are in [Appendix A.2](#).

The results are in [Table 6](#). LLMs are not good at counting, so we did not constrain the percentage length (i.e., sparsity) of the rationale extracted by the model. Comparing the results of the supervised fine-tuned llama-3.1 with our MGR-VRM, LLaMA-3.1 does not show a clear advantage. We do not claim that our method can outperform other powerful (but expensive) LLMs, as stronger LLMs can always be trained with more computational resources and greater human effort. Nevertheless, considering that our focus is interpretability, that our model does not rely on manually annotated rationale labels, and that it operates with low consumption cost, our research and findings remain valuable.

7. Conclusion

In this paper, we introduce a novel method for extracting causal rationales from a probabilistic standpoint, without requiring graphical causal models and their associated assumptions. Our approach demonstrates significantly enhanced performance, while using computational resources comparable to the latest causal methods. This work contributes to two key areas: interpretability and data cleaning. First, we introduce causal explanations, providing a more reliable basis for decision-making compared to traditional associative explanations. Additionally, the data-centric causal relationships (i.e., model-agnostic patterns inherent in the dataset) can potentially be employed for data cleaning, thereby improving the quality of data used for fine-tuning large models.

CRedit authorship contribution statement

Wei Liu: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Zhiying Deng:** Writing – original draft, Methodology, Investigation; **Lang Gao:** Visualization, Validation, Software; **Yuankai Zhang:** Visualization, Validation, Software; **Haozhao Wang:** Writing – review & editing, Writing – original draft; **Jun Wang:** Writing – review & editing,

Supervision; **Rui Zhang**: Writing – review & editing, Investigation; **Heng Huang**: Writing – review & editing, Supervision; **Ruixuan Li**: Writing – review & editing, Supervision, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT-4o and DeepL Write in order to make the English presentation better. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data is provided with a link in our code.

Appendix A. More results

A.1. More implementation details

The statistics of the datasets are in [Table A.8](#).

We found that Inter-RAT is very sensitive to hyperparameters, so we keep the same settings as those used in Inter-RAT (i.e., 100-dimensional GloVe and 200-dimension GRUs) to avoid hyperparameters adversely affecting its results. Some baseline results are copied from previous papers.

For Inter-RAT, the hyperparameters are initialized using the values reported in their source codes and are then manually tuned multiple times to determine the optimal settings. Specifically, we directly use its reported results for the beer-related datasets. We use a batch size of 256 and a learning rate of 0.001 for the hotel-related datasets.

For RNP, Noise-RAT, GRAT, and our MGR-VRM, we find that they are variants of RNP, so we first tune the hyperparameters for RNP. According to some previous methods in this field, we find that a learning rate of 0.0001 works well for RNP on these

Table A.8
Statistics of datasets used in this paper.

Datasets		Train		Dev		Annotation		
		Pos	Neg	Pos	Neg	Pos	Neg	Sparsity
Beer	Appearance	202385	12897	28488	1318	923	13	18.5
	Aroma	172299	30564	24494	3396	848	29	15.6
	Palate	176038	27639	24837	3203	785	20	12.4
Hotel	Location	7236	7236	906	906	104	96	8.5
	Service	50742	50742	6344	6344	101	99	11.5
	Cleanliness	75049	75049	9382	9382	99	101	8.9

Table A.9
Hyperparameters for different methods.

Methods	Learning rate	Batch size
RNP	0.0001	128
Inter-RAT	0.001	256
Noise-RAT	0.0001	128
GRAT	0.0001	128
MGR	0.0001	128
MGR-VRM	0.0001	128

two datasets. Then, we search for the batch size within [64, 128, 256, 512], and we find that 128 is generally good for RNP. Since Noise-RAT and our MGR-VRM are just variants built on top of RNP, we adopt the learning rate and batch size of RNP. For Noise-RAT, another important hyperparameter is the ratio (p) of the injected noise. We search p from [0.1, 0.2, 0.3], and find that $p = .3$ is generally better. We list the hyperparameters in Table A.9. Other hyperparameters that are not explicitly mentioned (such as the temperature of Gumbel-Softmax and the weights for sparsity and coherence) are kept at their default values, consistent with prior literature. These hyperparameters are available in our code repository.

A.2. Details of experiments with llama

To further show the potential impact of rationalization in the era of LLMs, we present experimental results using the LLaMA-3.1-8B-Instruct model. We perform both two-shot prompting and supervised fine-tuning.

For two-shot prompting, we provide the model with one negative text with its corresponding rationale, and one positive text with its corresponding human-annotated rationale. For supervised fine-tuning, the supervision signal is the classification label since we perform unsupervised rationale extraction. We use LoRA to fine-tune the models. We provide a detailed document in our code repository to include all the details (including the prompt templates, LoRA fine-tuning parameter settings, and more).

In most cases, the model can output the rationale in the correct format. Fig. A.7 shows an example. But in two-shot prompting, the model sometimes outputs additional parts along with the rationale (through manual observation, this situation does not occur frequently.). Fig. A.6 is another example. In such cases, we use GPT-3.5-turbo to extract the content within the quotation marks.

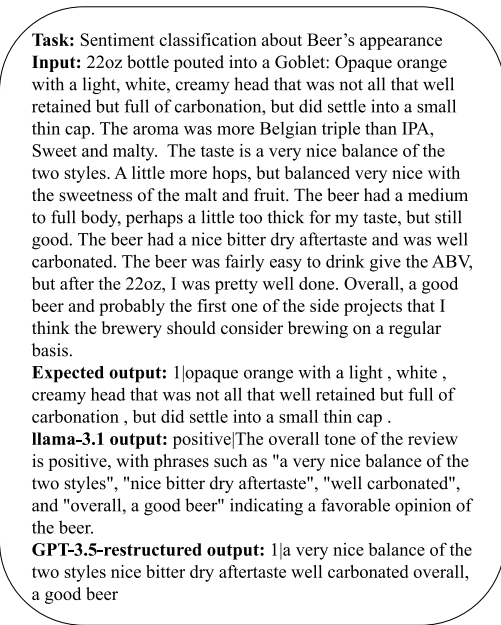


Fig. A.6. An example of llama fails to output the rationale in the right format.

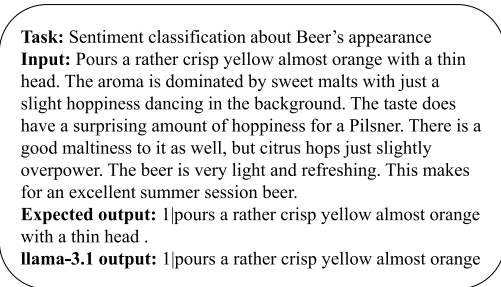


Fig. A.7. An example of llama's output. Here, "1" means that the class label Y is positive. The words after "|" represent the rationale.

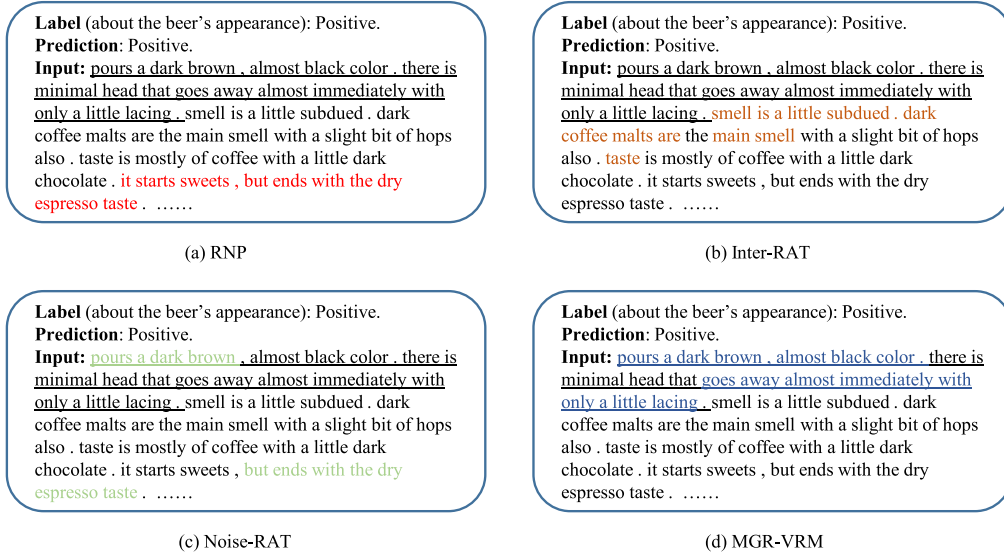


Fig. A.8. An example of selected rationales in the Beer-Appearance dataset. The sparsity is set to be about 10%. The underlined texts are human-annotated (causal) rationales. (a): RNP selects taste only. (b): Inter-RAT selects smell. (c): Noise-RAT selects both appearance and taste. (d): MGR-VRM selects appearance only.

A.3. Visualized examples

We provide a practical example of the rationales selected by different methods in Fig. A.8. Both RNP and Inter-RAT fail to extract the causal rationale about the beer’s appearance. Although Noise-RAT selects part of the causal rationale, it also selects the comments about the beer’s taste. Our MGR-VRM successfully selects the causal rationale.

A.4. Results with more generator heads

In the main experiments, we fix the number of generators to $n = 3$ to evaluate the effectiveness of MGR/MGR-VRM in comparison with other baseline methods. We additionally report results obtained with a larger number of generators. The experiments are conducted on the Beer-Aroma dataset, with the sparsity level set to approximately 20%. For different generators in MGR-VRM, we always set $S_i \approx i \times 10\%$. The results are summarized in Table A.10. Both MGR and MGR-VRM already achieve strong performance with only $n = 3$ generators, outperforming all baseline methods. While increasing n further continues to improve performance, the marginal gain from each additional generator is smaller than the per-generator gain observed at $n = 3$. Therefore, we adopt $n = 3$ in our main experiments.

A.5. Results of sharing different layers

In our main experiments, MGR-VRM shares the entire encoder across generators. Here, we present a fine-grained analysis by examining the results of sharing different subsets of layers. We use BERT-base-uncased as the encoder (12 layers in total) and follow the experimental settings corresponding to Table 4. Specifically, we report results for sharing the first 3 layers, the first 6 layers, the first 9 layers, and all 12 layers of the encoder. The results are shown in Table A.11. The results indicate that the performance of MGR-VRM remains stable across different layer-sharing configurations, suggesting that the method is robust to the degree of encoder sharing.

Table A.10

Results of MGR with different numbers of generators. The dataset is Beer-Aroma. The sparsity is $S \approx 20$.

Beer-Aroma	MGR				MGR-VRM			
	S	P	R	F1	S	P	R	F1
$n = 1$ (RNP)	20.0	37.5	51.9	43.5	20.0	37.5	51.9	43.5
$n = 3$ (MGR)	19.7	64.4	81.3	71.9	19.2	69.3	85.6	76.6
$n = 5$	19.5	67.3	82.7	74.2	19.5	70.0	87.7	77.9
$n = 7$	19.6	68.0	83.6	75.0	19.5	70.5	88.0	78.3
$n = 9$	19.6	67.9	83.7	75.0	19.6	70.3	88.5	78.4

Table A.11
Results of MGR-VRM under different shared encoder layers.

Methods	Beer-Appearance			
	S	P	R	F1
3 layers	11.0	82.3	38.4	52.5
6 layers	11.2	82.2	38.9	52.9
9 layers	11.4	81.4	38.8	52.6
12 layers	11.3	82.0	38.9	52.8

Appendix B. Technical proofs

B.1. The relation between entropy and cross-entropy

It is a basic idea in information theory that the entropy of a distribution P is upper bounded by the cross-entropy of using Q to approximate it. For any two distributions P and Q , we have

$$H_c(P, Q) = H(P) + D_{KL}(P||Q) \geq H(P), \quad (\text{B.1})$$

where the subscript c in $H_c(P, Q)$ stands for cross-entropy.

We know that we get the minimum cross-entropy when Q is the same as P , i.e., $D_{KL}(P||Q) = 0$. Which means that

$$\min H_c(P, Q) = H(P). \quad (\text{B.2})$$

B.2. Proof of Lemma 2

Definition 1 (MGF). Let $X \in \mathbb{R}$ be a random variable, the function

$$M(s) = E(\exp(sX)) \quad (\text{B.3})$$

is called the moment generating function (MGF) of X .

Remark 5. For $X \sim N(\mu, \sigma^2)$, its MGF is $M(s) = E(\exp(sX)) = \exp(s\mu + \frac{\sigma^2 s^2}{2})$.

Lemma 6. Let $X \sim N(\mu, \sigma^2)$, $\forall s, t > 0$, we have

$$\begin{aligned} \Pr(X - \mu > t) &= \Pr(\exp(sX - s\mu) > \exp(st)) \\ &\leq \frac{E(\exp(sX - s\mu))}{\exp(st)} \\ &= \exp(-st + \frac{\sigma^2 s^2}{2}). \end{aligned} \quad (\text{B.4})$$

The above “ $\leq$ ” is from the Chebyshev inequality. By setting $s = \frac{t}{\sigma^2}$, we then have $\Pr(X - \mu > t) \leq \exp(-\frac{t^2}{2\sigma^2})$. Utilizing the symmetry of the normal distribution, we then easily get Lemma 2.

B.3. Proof of Lemma 3

To make the presentation succinct, we first discuss the case where n is an odd number.

$$\begin{aligned} &\lim_{n \rightarrow \infty} \Pr(K < n - K) \\ &= \lim_{n \rightarrow \infty} \Pr(K < \frac{n}{2}). \end{aligned} \quad (\text{B.5})$$

We denote $\Phi(x)$ as the cumulative distribution function of $X \sim N(0, 1)$. Using Lemma 1, we have that

$$\Pr(\frac{K - np}{\sqrt{np(1-p)}} \leq x) \rightarrow \Phi(x), \text{ s.t. } n \rightarrow \infty. \quad (\text{B.6})$$

Then, we have

$$\Pr(K < \frac{n}{2}) = \Pr(\frac{K - np}{\sqrt{np(1-p)}} < \frac{\frac{n}{2} - np}{\sqrt{np(1-p)}}) \rightarrow \Phi(c\sqrt{n}), \quad (\text{B.7})$$

where $c = \frac{0.5-p}{\sqrt{p(1-p)}} < 0$ is a fixed value.

Using Lemma 2, we have

$$\Phi(x) = \Pr(X \leq x) \leq \exp(-\frac{1}{2}x^2), \text{ s.t. } \forall x < 0. \quad (\text{B.8})$$

So, $\Phi(c\sqrt{n}) \leq \exp(-\frac{1}{2}c^2n)$. Then,

$$\lim_{n \rightarrow \infty} \Pr(K < \frac{n}{2}) \leq \lim_{n \rightarrow \infty} \exp(-\frac{1}{2}c^2n) = 0 \quad (\text{B.9})$$

There is nothing different except that $\Pr(K < \frac{n}{2})$ should be replaced by $\Pr(K < \frac{n}{2} - 1)$ when n is an even number.

B.4. Proof of Lemma 4

According to Eq. (7), we have

$$\begin{aligned} \frac{\partial R_P(\alpha)}{\partial \alpha} &= a \cdot K - (n - K)a - K \cdot b + (n - K)b \\ &= a \cdot (K - (n - K)) - b \cdot (K - (n - K)) \\ &= (a - b)(K - (n - K)). \end{aligned} \quad (\text{B.10})$$

Since we have $a > b$ and $K > n - K$, we get $\frac{\partial R_P(\alpha)}{\partial \alpha} > 0$. It means that to get a higher payoff, the predictor needs to increase α , i.e., it needs to move towards θ_p^1 . The proof of Lemma 4 is completed.

B.5. Proof of Theorem 1

Based on the derivation of Lemma 3, the proof now becomes obvious. Eq. (13) is just equal to $\lim_{n \rightarrow \infty} \Pr(K < n - K) = 0$. Combining Eqs. (B.7) and (B.5), we know $\Pr(K < n - K)$ is a monotonically decreasing function of n , which leads to Eq. (14).

B.6. Proof of Lemma 5

$$\begin{aligned} & D_{KL}(P_{Y|A} || P_{\hat{Y}|A}) - D_{KL}(P_Y || P_{\hat{Y}}) \\ &= \sum_{a \in \mathcal{A}} \sum_{y \in \mathcal{Y}} P_{Y,A}(y, a) \cdot \ln \frac{P_{Y|A}(y|a)}{P_{\hat{Y}|A}(y|a)} - \sum_{y \in \mathcal{Y}} P_Y(y) \cdot \ln \frac{P_Y(y)}{P_{\hat{Y}}(y)} \\ &= \sum_{a \in \mathcal{A}} \sum_{y \in \mathcal{Y}} P_{Y,A}(y, a) \cdot \ln \frac{P_{Y|A}(y|a)}{P_{\hat{Y}|A}(y|a)} \\ &\quad - \sum_{y \in \mathcal{Y}} (\sum_{a \in \mathcal{A}} P_{Y,A}(y, a)) \cdot \ln \frac{P_Y(y)}{P_{\hat{Y}}(y)} \\ &= \sum_{a \in \mathcal{A}} \sum_{y \in \mathcal{Y}} P_{Y,A}(y, a) \cdot \ln \frac{P_{Y|A}(y|a)P_{\hat{Y}}(y)}{P_{\hat{Y}|A}(y|a)P_Y(y)} \\ &\geq \sum_{a \in \mathcal{A}} \sum_{y \in \mathcal{Y}} P_{Y,A}(y, a) (1 - \frac{P_{\hat{Y}|A}(y|a)P_Y(y)}{P_{Y|A}(y|a)P_{\hat{Y}}(y)}) \\ &= 1 - \sum_{y \in \mathcal{Y}} (\frac{P_Y(y)}{P_{\hat{Y}}(y)} \sum_{a \in \mathcal{A}} P_A(a)P_{\hat{Y}|A}(y|a)) \\ &= 1 - \sum_{y \in \mathcal{Y}} \frac{P_Y(y)}{P_{\hat{Y}}(y)} P_{\hat{Y}}(y) \\ &= 0. \end{aligned} \quad (\text{B.11})$$

(*) is from $\ln x \geq 1 - \frac{1}{x}$.

B.7. Discussion about $P_c > 0.5$

1) Intuitive analysis and empirical support.

For a dataset, there are some samples that contain both the causality and the spurious correlation (i.e., X_1 and X_2 , corresponding to the number of Correlated in Table B.12), and the other samples contain only the causality (i.e., X_1 , corresponding to the number of Decorrelated in Table B.12). So we always have the number of X_1 is larger than that of X_2 . For random selection, the probability of selecting X_1 is higher than selecting X_2 , which means that $P_c > 0.5$. In Table B.12, we approximate P_c by

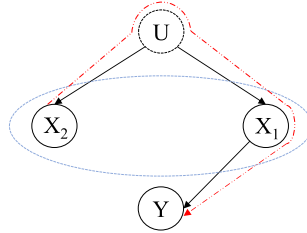
$$\begin{aligned} P_c &= \frac{\text{Number}(X_1)}{\text{Number}(X_1) + \text{Number}(X_2)} \\ &= \frac{(\text{Decorrelated} + \text{Correlated})}{(\text{Decorrelated} + \text{Correlated}) + \text{Correlated}} > 0.5 \\ &= \frac{15169 + 15395}{15169 + 15395 * 2} = 0.67. \end{aligned} \quad (\text{B.12})$$

Table B.12

The P_c approximated by the statistical data of the *Beer-Aroma* dataset. The statistics are collected by Lei et al. [9]. It is approximated by

$$1 - \frac{\text{Correlated}}{\text{Correlated} * 2 + \text{Decorrelated}}.$$

Full	Decorrelated	Correlated	P_c
30564	15169	15395	0.67

**Fig. B.9.** The graphical model for the data-generating process.

We only count samples with negative labels because the original dataset is unbalanced and we do sampling balance according to the number of negative samples during training.

2) Formal analysis with probabilistic graphical models.

To begin with, we want to clarify that although we have posited a probabilistic graphical model here, we are not using it to do the causal analysis. Instead, we use it only to show why $P_c > 0.5$, which is testable.

Formally, we posit a probabilistic graphical model to illustrate the corresponding data-generating process, which is shown in Fig. B.9. The annotators assign the label Y by viewing the causal rationale X_1 ($X_1 \rightarrow Y$). X_2 and X_1 are correlated through some unobservable confounders U . For example, U may include a variable indicating whether the beer originates from a reputable brand, and X_1, X_2 denote the comments about the beer smell and taste respectively. Y is the sentiment label about the beer's smell. A pleasant taste may imply that the beer comes from a good brand ($U \rightarrow X_2$). Moreover, a beer from a reputable brand is likely to have a pleasing smell ($U \rightarrow X_1$). Consequently, X_2 is associated with Y via a *backdoor* path, as depicted by the red dotted line. Without losing generality, we consider that U, X_1, X_2, Y are all Bernoulli variables with $B(1, 0.5)$. We assume a conditional probability for each directed arrow:

$$\begin{aligned} \Pr(X_1 = 1|U = 1) &= \Pr(X_1 = 0|U = 0) = p_{UX_1} > 0.5, \\ \Pr(X_2 = 1|U = 1) &= \Pr(X_2 = 0|U = 0) = p_{UX_2} > 0.5, \\ \Pr(Y = 1|X_1 = 1) &= \Pr(Y = 0|X_1 = 0) = p_{X_1Y} > 0.5, \end{aligned} \quad (\text{B.13})$$

where “ > 0.5 ” means positive causation here. In Fig. B.9, we have $X_1 \perp\!\!\!\perp X_2|U$. Then based on Bayes' theorem and the Law of Total Probability, we can easily get

$$\begin{aligned} p_{X_2Y} &= \Pr(Y = 1|X_2 = 1) \\ &= p_{X_1Y} + (2p_{UX_1}p_{UX_2} - p_{UX_1} - p_{UX_2})(2p_{X_1Y} - 1). \end{aligned} \quad (\text{B.14})$$

Since $2p_{X_1Y} - 1 > 0$ and $2p_{UX_1}p_{UX_2} - p_{UX_1} - p_{UX_2} < 0$, we have $p_{X_2Y} < p_{X_1Y}$. Then we have

$$P_c = \frac{p_{X_1Y}}{p_{X_1Y} + p_{X_2Y}} > 0.5. \quad (\text{B.15})$$

B.8. Generalizing Theorem 1 to general cases

Here we consider X consists of X_1, X_2, X_3 , where X_1 is the causal rationale and X_2 and X_3 are two different spurious correlations. The core derivation process of Theorem 1 can be summarized as follows: if the probability of occurrence for X_1 in the dataset is higher than that of X_2 , then a randomly initialized generator has a higher probability of selecting X_1 . Based on this, we can increase the number of generators to raise the probability of having more generators select X_1 than X_2 . With more generators selecting X_1 , the predictor will optimize towards fitting X_1 , further increasing the probability of X_1 being selected by the next round of generators.

Expanding this to the case where we have $\langle X_1, X_2, X_3 \rangle$, we only need to consider the cases of $\langle X_1, X_2 \rangle$ and $\langle X_1, X_3 \rangle$, and the derivation process is almost identical. Also, there is nothing different for the case where we have $\langle X_1, X_2, \dots, X_n \rangle$.

B.9. Proof of Theorem 2

We first prove the left inequality of Theorem 2. For any two random variable Z_i, Z_j , we have

$$\begin{aligned} H(Z_i|Z_j) &= H(Z_i, Z_j) - H(Z_j) \\ H(Z_i|Z_j) &\geq 0 \\ H(Z_j) &\geq 0. \end{aligned} \tag{B.16}$$

So, we have

$$H(Z_i, Z_j) \geq H(Z_j), \tag{B.17}$$

where the equality holds if and only if $H(Z_i|Z_j) = 0$, i.e., $Z_i = Z_j$. There is nothing different for $H(Z_i, Z_j) \geq H(Z_i)$. Then we easily get the left inequality of Theorem 2 through Mathematical Induction.

Then we prove the right inequality of Theorem 2. We first have

$$\begin{aligned} I(Z_i, Z_j) &= H(Z_i) + H(Z_j) - H(Z_i, Z_j) \\ I(Z_i, Z_j) &\geq 0, \end{aligned} \tag{B.18}$$

where $I(Z_i, Z_j)$ is the mutual information. $I(Z_i, Z_j) = 0$ if and only if $Z_i \perp\!\!\!\perp Z_j$. So, we have

$$H(Z_i, Z_j) \leq H(Z_i) + H(Z_j), \tag{B.19}$$

with equality holding if and only if $Z_i \perp\!\!\!\perp Z_j$. Then we easily get the right inequality of Theorem 2 through mathematical induction.

The proof of Theorem 2 is completed.

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